EX.No.2.1 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

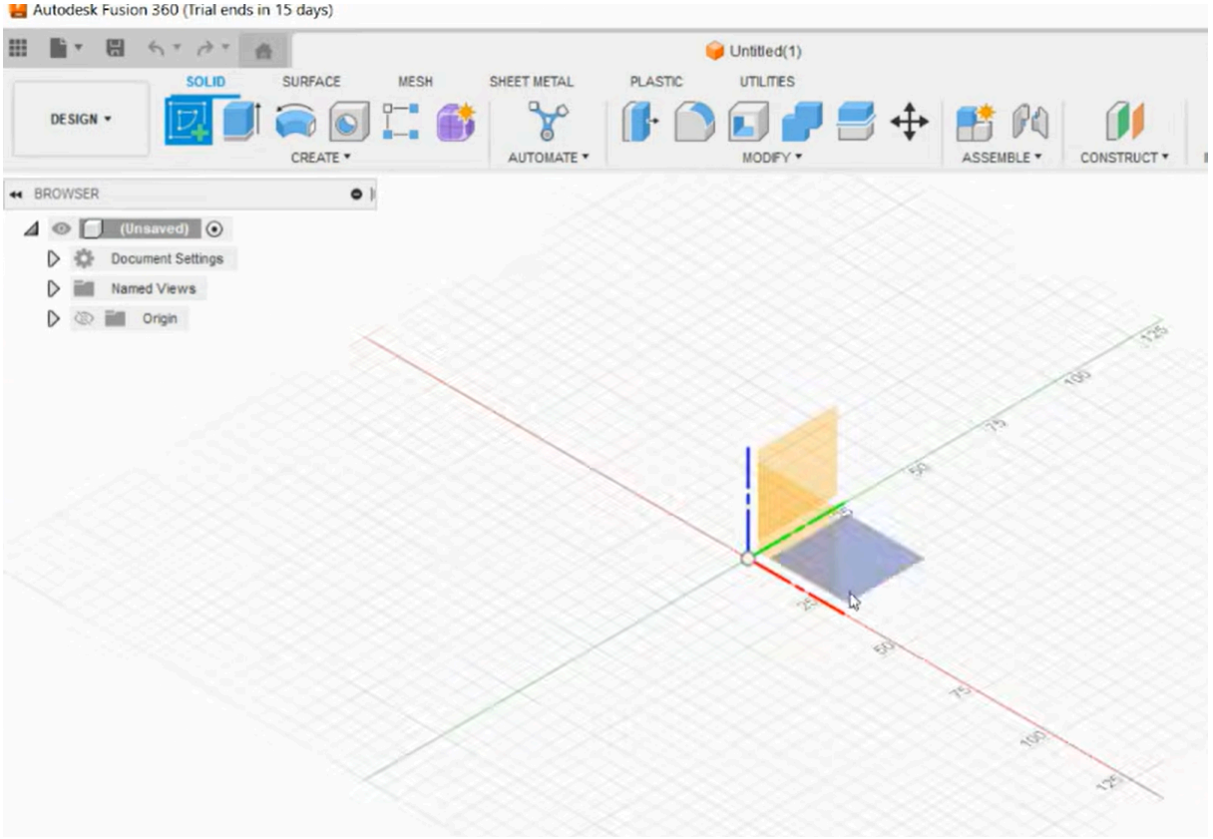
AIM:

To create a 3Dmodel from the given isometric drawing using fusion 360 tool.

REQUIREMENTS:

1.Autodesk fusion 360
2.Windows 10
3.16 GB of RAM (integrated graphics recommend 1 GB or more)
4.2.5 Mbps or faster download; 500 Kbps or faster upload

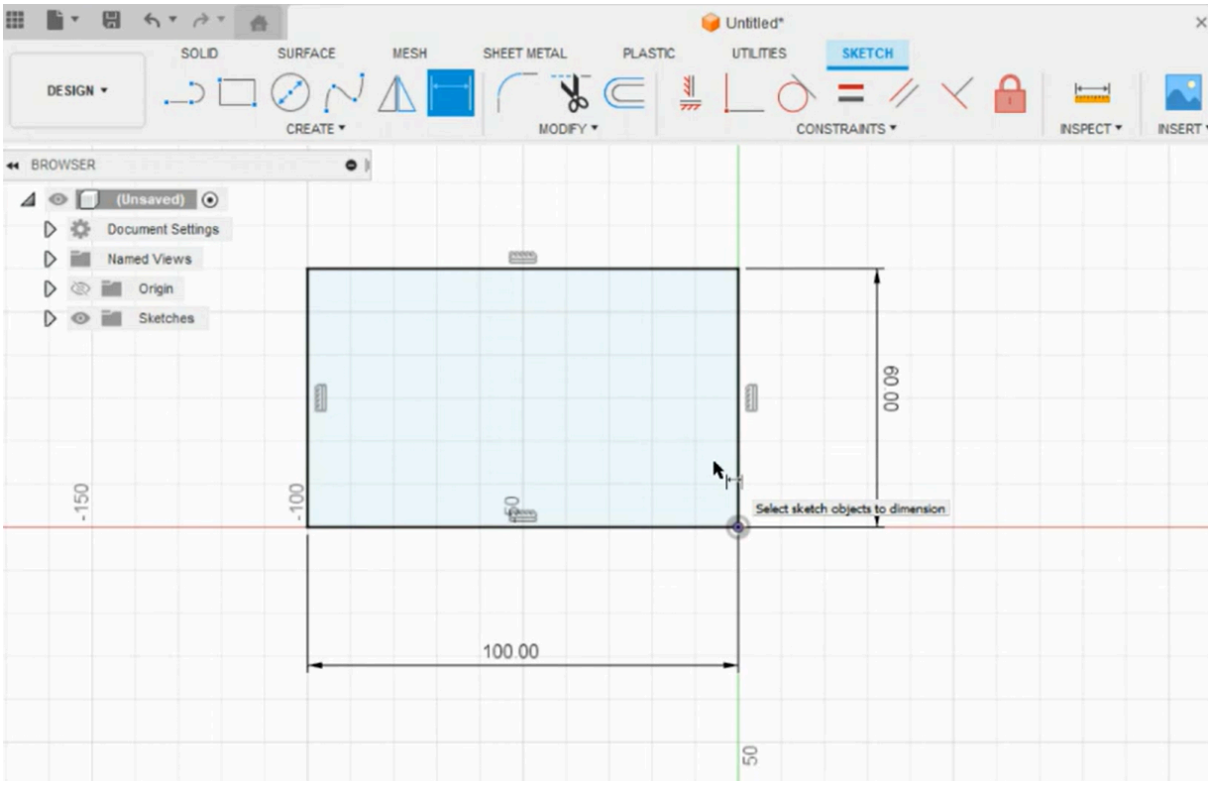
PROCEDURE:
1.First we should open the fusion 360,then select new to create a new project.
2.We should select the view as front or top or which view we want.
3.Then click create sketch to start our project.



4.Select the Extrude tool and enter a height of 50 mm to create the main rectangular block.

5. Select the front face of the block and create a new sketch. Draw a circle with a diameter of 50 mm at the required position.

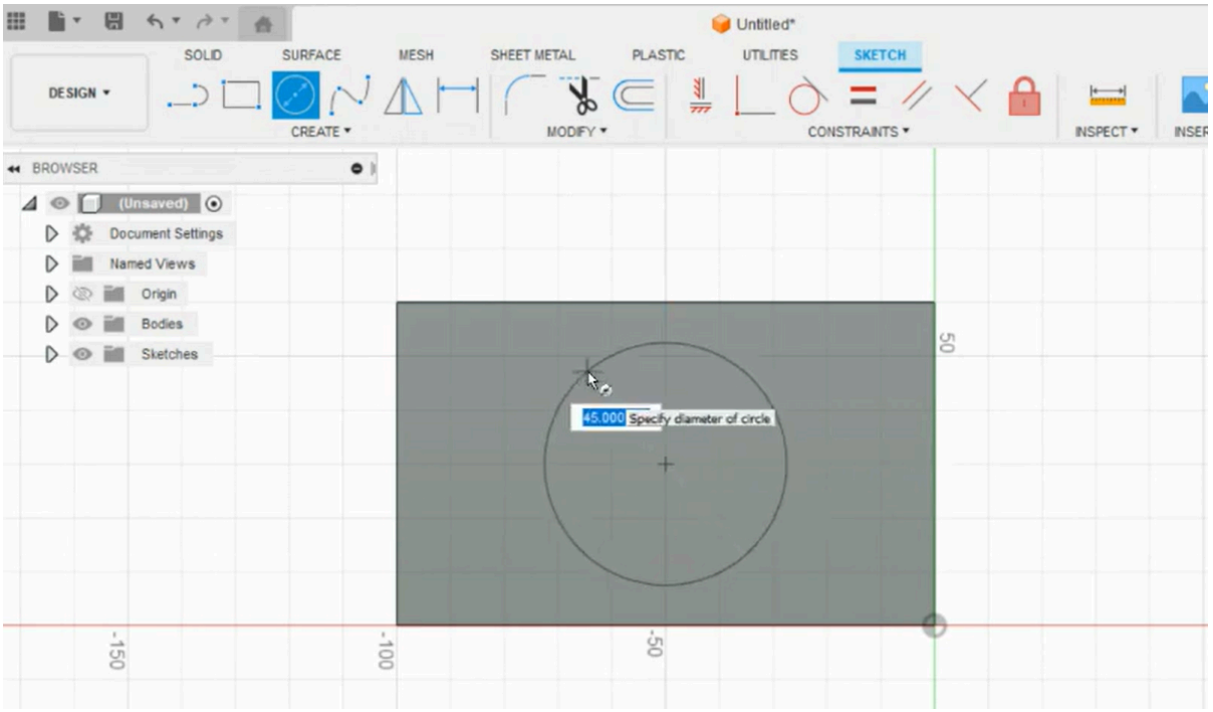
6. Select the Extrude tool and extend the circular profile outward by 20 mm to create the front cylindrical projection.



7.Select the front face of the block and create a new sketch. Draw a circle with a diameter of 50 mm at the required position.

8. Select the Extrude tool and extend the circular profile outward by 20 mm to create the front cylindrical projection.

9. Select the top face of the main block and create a new sketch. Draw a circle with a diameter of 20 mm at the specified position.

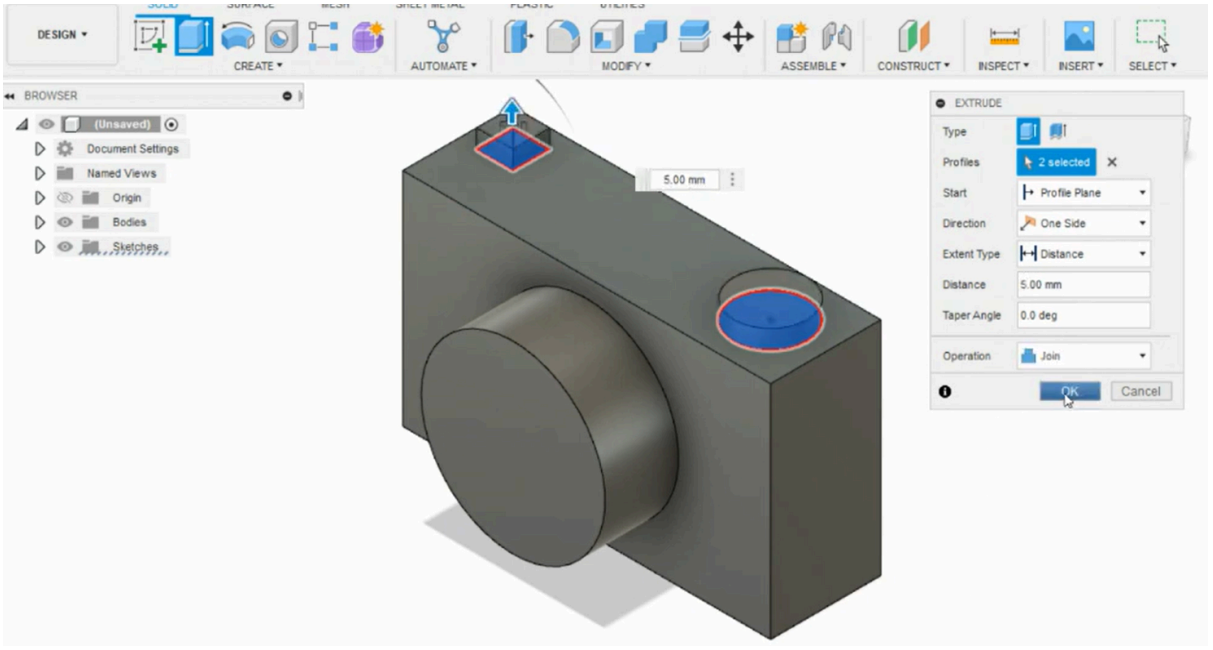


10.Extrude the circular profile upward by 10 mm to create the top cylindrical projection.

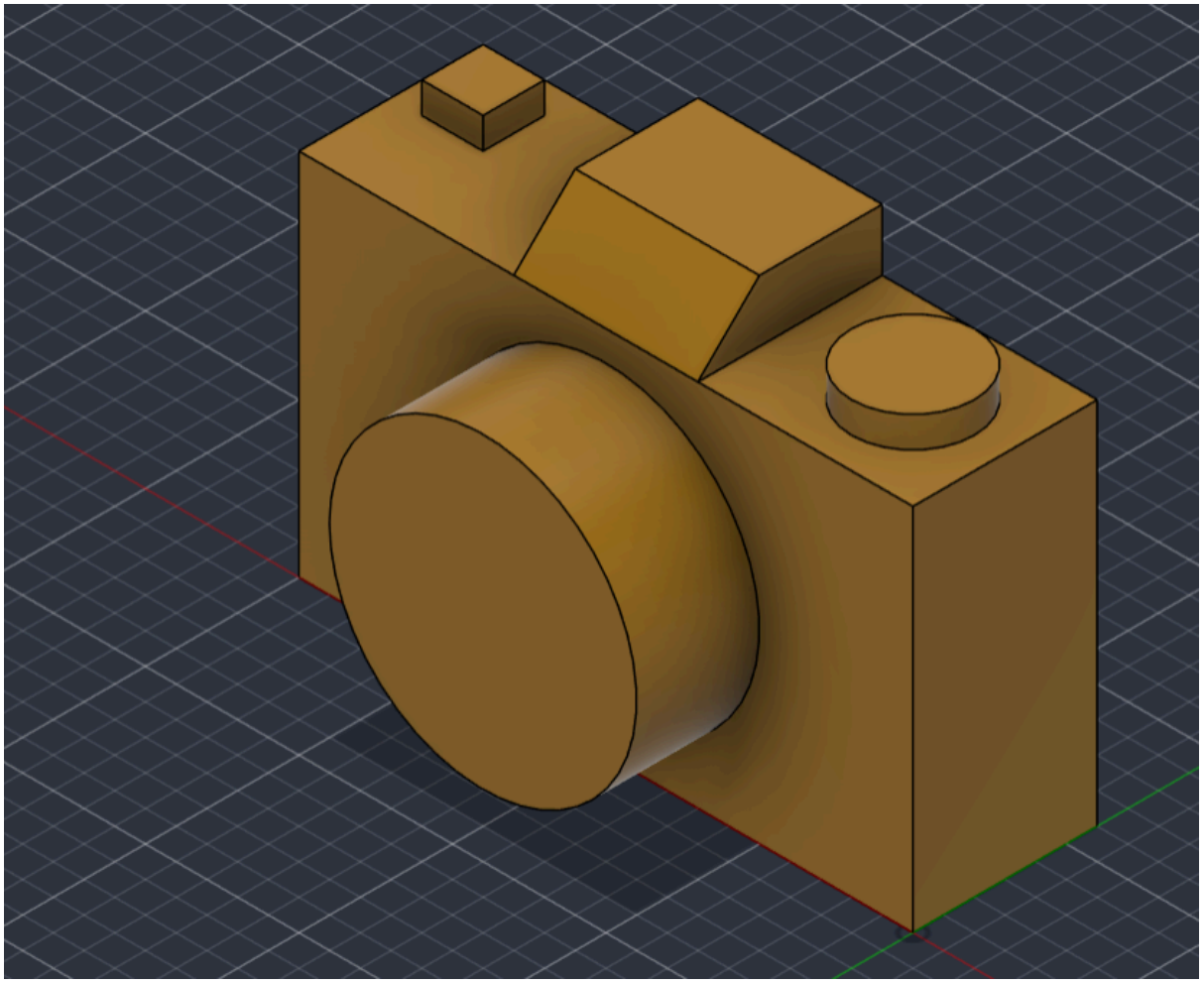
11. Create a sketch on the required plane and use the Line tool to draw the upper raised portion according to the given inclined shape and 45° angle.

12. Use the Extrude tool to give the required width to the raised portion and create the upper sloping block.

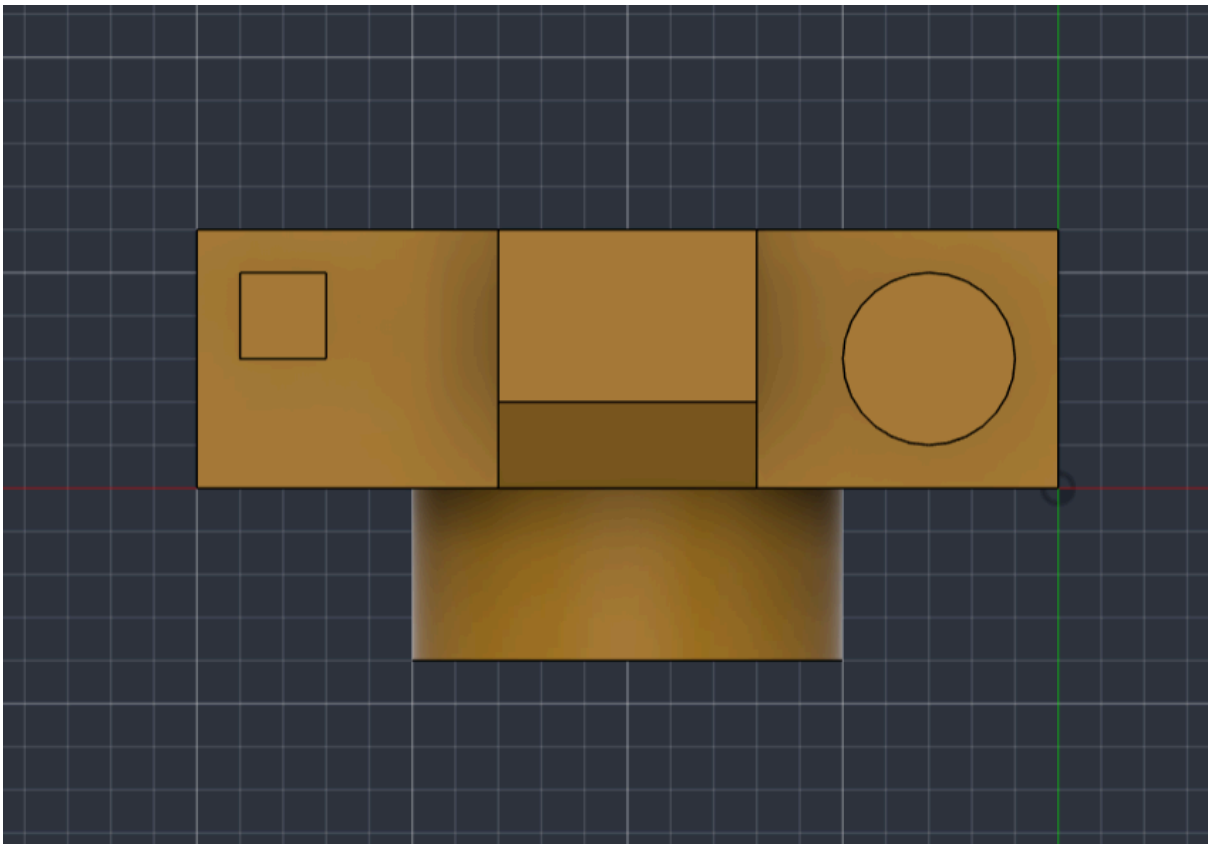
13. Select the top surface near the rear side and create a small rectangular sketch with the dimensions shown in the reference drawing.



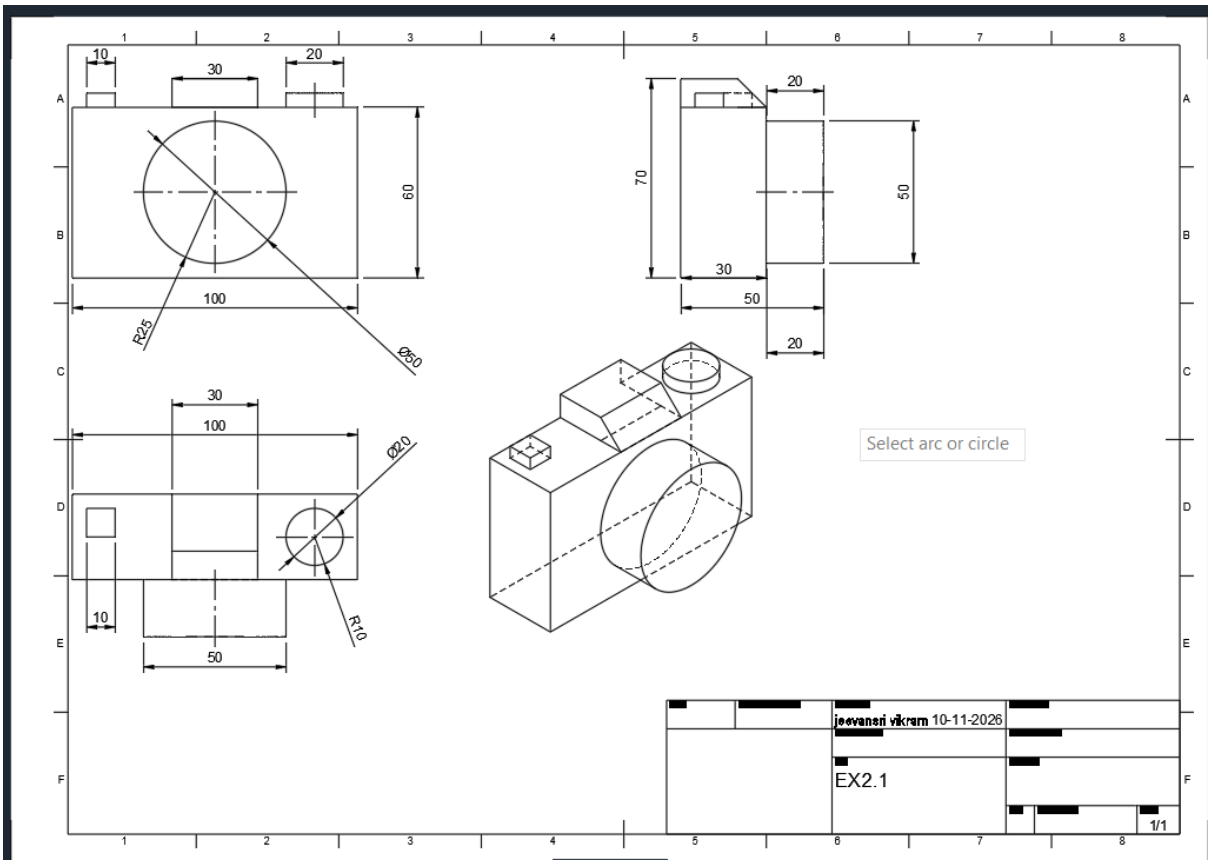
14. So finally we conclude our project by extrude it overall by the given mm.



SKETCHES:



OUTPUT:



RESULT:

Thus, the 3D model is created from given isometric drawing.